

A Map Between Arborifications of Multiple Zeta Values

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- ① Multiple zeta values
- ② The arborifications
- ③ Manchon's question (2020)
- ④ Main result
- ⑤ References

Multiple zeta values

Multiple zeta values(MZVs)

Definition (Multiple zeta values)

- $k_1, \dots, k_d \in \mathbb{N}, k_d > 1$

$$\zeta(k_1, \dots, k_d) := \sum_{0 < n_1 < \dots < n_d} \prod_{i=1}^d \frac{1}{n_i^{k_i}}$$

Definition (Iterated integral)

- $a_0, a_1, \dots, a_k, a_{k+1} \in \{0, 1\}$
- $\omega_0(t) := \frac{dt}{t}, \omega_1(t) := \frac{dt}{t-1}$

$$I(a_0; a_1, \dots, a_k; a_{k+1}) := \int_{a_0 < t_1 < \dots < t_k < a_{k+1}} \prod_{j=1}^k \omega_{a_j}(t_j)$$

Remark (iterated integral expression of MZVs)

$$\zeta(k_1, \dots, k_d) = (-1)^d I(0; 1, \{0\}^{k_1-1}, \dots, 1, \{0\}^{k_d-1}; 1)$$

Hopf algebra structure

Definition (Hopf algebra for MZVs)

- $\mathcal{X} := \{x_0, x_1\}$
- $\mathbb{Q}\langle\mathcal{X}\rangle :=$ non-commutative polynomial algebra generated by $\mathcal{X}$
 $(\mathbb{Q}\langle\mathcal{X}\rangle, \sqcup, \Delta) : \text{Hopf algebra with shuffle product for integral}$
- $\mathcal{Y} := \{y_n \mid n \in \mathbb{N}\}$
- $\mathbb{Q}\langle\mathcal{Y}\rangle :=$ non-commutative polynomial algebra generated by $\mathcal{Y}$
 $(\mathbb{Q}\langle\mathcal{Y}\rangle, *, \Delta) : \text{Hopf algebra with stuffle product for series}$

Remark (correspondence between integral and series)

$$\begin{array}{ccc}
 y_{n_1} \cdots y_{n_r} & \xrightarrow{s} & x_1 x_0^{n_1-1} \cdots x_1 x_0^{n_r-1} \\
 \downarrow & & \downarrow \\
 \sum_{0 < m_1 < \cdots < m_r} \prod_{i=1}^r \frac{1}{m_i^{n_i}} & & I(0; 1, \{0\}^{n_1-1}, \dots, 1, \{0\}^{n_r-1}; 1)
 \end{array}$$

The arborifications

The arborifications

Example

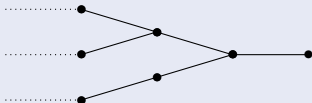
- 1 total order

$$x_1 \prec x_2 \prec x_3 \prec \cdots \prec x_{n-2} \prec x_{n-1} \prec x_n$$

- ② ladder tree



- ③ rooted tree



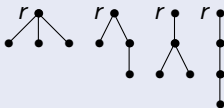
- ④ partial order

$$\begin{array}{ccccccc} \cdots & \curvearrowright & X_e & & \curvearrowright & X_c & \\ \cdots & \curvearrowright & X_f & & \curvearrowright & & \\ \cdots & \curvearrowright & X_g & & \curvearrowright & X_d & \\ & & & & & \curvearrowright & X_b \curvearrowright X_a \end{array}$$

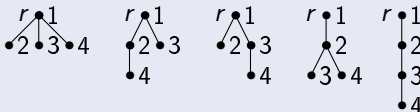
Rooted trees

Example (Rooted trees)

An example of (non-planar) **rooted tree** with $|V(T)| = 4$



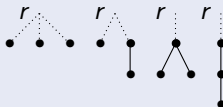
An example of **planar rooted tree** with $|V(T)| = 4$



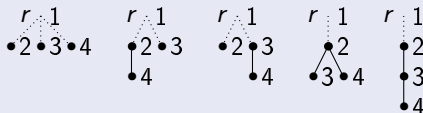
Rooted forests

Example (Rooted forests)

An example of (non-planar) **rooted forest** with $|V(T)| = 3$



An example of **planar rooted forest** with $|V(T)| = 3$

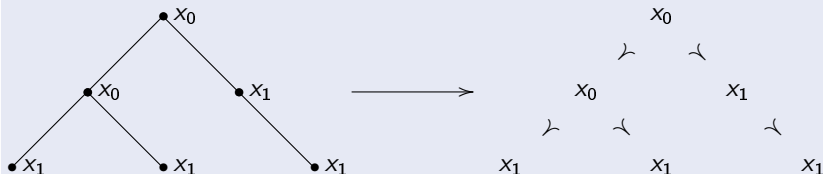


$\mathcal{D}$ -decoration

Definition

Let $\mathcal{D}$ be a set. A **$\mathcal{D}$ -decorated rooted tree (resp. forest)** is a rooted tree D (resp. forest F_D) together with a decoration map δ_D (resp. δ_{F_D}) from the vertex set $V(D)$ (resp. $V(F_D)$) to $\mathcal{D}$, where $\mathcal{D}$ is $\mathcal{X} := \{x_0, x_1\}$, $\mathcal{Y} := \{y_n \mid n \in \mathbb{N}\}$.

Example ($\mathcal{X}$ -decorated rooted tree with opposite tree order)



Arborified multiple zeta values of the first kind

Definition

Arborified multiple zeta values of the first kind are multiple zeta values associated with a $\mathcal{Y}$ -decorated rooted tree Y (resp. forest), defined as the harmonic series associated to the triple $(V(Y), \preceq_Y, \delta_Y)$.

$$\zeta(Y) := \sum_{\substack{n_v \in \mathbb{N} \\ n_u < n_v \text{ if } u \prec_Y v}} \prod_{v \in V(Y)} \frac{1}{n_v^{k_v}},$$

where k_v is the integer n such that $\delta_Y(v) = y_n$.

Example

$$\zeta \left(\begin{array}{c} \bullet \\ / \quad \backslash \\ \bullet \quad \bullet \\ y_1 \quad y_3 \\ \backslash \quad / \\ \bullet \\ y_2 \end{array} \right) = \zeta(1, 3, 2) + \zeta(3, 1, 2) + \zeta(4, 2)$$

Arborified multiple zeta values of the second kind

Definition

Arborified multiple zeta values of the second kind are multiple zeta values associated with a $\mathcal{X}$ -decorated rooted tree X (resp. forest), defined as Yamamoto's integral associated to the triple $(V(X), \preceq_X, \delta_X)$.

$$\zeta(X) := I(X) = \int_{\Delta(X)} \prod_{v \in V(X)} \omega_{\delta_X(v)}(t_v),$$

where $\Delta(X) := \{t = (t_v)_{v \in V(X)} \in (0, 1)^{V(X)} \mid t_u < t_v \text{ if } u \prec_X v\}$,
 $\omega_{x_0}(t) := \frac{dt}{t}$, $\omega_{x_1}(t) := \frac{dt}{1-t}$.

Example

$$\zeta \left(\begin{array}{c} \bullet \\ \swarrow \quad \searrow \\ \bullet \quad \bullet \\ x_1 \quad x_1 \end{array} \right) = I(0; 1, 1, 0; 1) + I(0; 1, 1, 0; 1) = 2\zeta(1, 2)$$

Hopf algebra structure

Definition (Butcher-Connes-Kreimer Hopf algebra)

- $\mathcal{D}$: set
- $\mathcal{T}^{\mathcal{D}}$ (resp. $\mathcal{T}^{PD}$) : set of non-empty $\mathcal{D}$ -decorated (resp. planar) rooted trees
- $\mathbb{Q}[\mathcal{T}^{\mathcal{D}}]$ (resp. $\mathbb{Q}\langle\mathcal{T}^{PD}\rangle$) : commutative (resp. non-commutative) polynomial algebra generated by $\mathcal{T}^{\mathcal{D}}$ (resp. $\mathcal{T}^{PD}$)
- $\mathcal{H}_{BCK}^{\mathcal{D}} := (\mathbb{Q}[\mathcal{T}^{\mathcal{D}}], \pi, \Delta)$ (resp. $\mathcal{H}_{NBCK}^{PD} := (\mathbb{Q}\langle\mathcal{T}^{PD}\rangle, \pi, \Delta)$) : (resp. non-commutative) BCK Hopf algebra of $\mathcal{D}$ -decorated (resp. planar) rooted tree

Example

$$\mathcal{H}_{BCK}^{\mathcal{X}} := (\mathbb{Q}[\mathcal{T}^{\mathcal{X}}], \sqcup, \Delta) \text{ (resp. } \mathcal{H}_{NBCK}^{PX} := (\mathbb{Q}\langle\mathcal{T}^{PX}\rangle, \sqcup, \Delta))$$

$$\mathcal{H}_{BCK}^{\mathcal{Y}} := (\mathbb{Q}[\mathcal{T}^{\mathcal{Y}}], *, \Delta) \text{ (resp. } \mathcal{H}_{NBCK}^{PY} := (\mathbb{Q}\langle\mathcal{T}^{PY}\rangle, *, \Delta))$$

Manchon's question (2020)

Manchon's question

Based on Foissy's work, Manchon introduced the simple arborification $\mathfrak{a}_{\mathcal{X}}$ and the contracting arborification $\mathfrak{a}_{\mathcal{Y}}$, which are Hopf algebra morphisms.

Question

Manchon posed the question to find a natural map $\mathfrak{s}^T$ with respect to the tree structures, which makes the following diagram commutative.

$$\begin{array}{ccc}
 \mathcal{H}_{BCK}^{\mathcal{Y}} & \xrightarrow{\mathfrak{s}^T} & \mathcal{H}_{BCK}^{\mathcal{X}} \\
 \mathfrak{a}_{\mathcal{Y}} \downarrow & & \downarrow \mathfrak{a}_{\mathcal{X}} \\
 \mathbb{Q}\langle \mathcal{Y} \rangle & \xrightarrow{\mathfrak{s}} & \mathbb{Q}\langle \mathcal{X} \rangle
 \end{array}$$

Manchon's question

Definition (ladder tree section)

$$\ell_{\mathcal{X}}(x_{m_1} x_{m_2} \cdots x_{m_s}) := \begin{array}{c} \bullet \quad x_{m_s} \\ \vdots \\ \bullet \quad x_{m_2} \\ | \\ \bullet \quad x_{m_1} \end{array} \quad \left(\text{resp. } \ell_{\mathcal{Y}}(y_{n_1} y_{n_2} \cdots y_{n_t}) := \begin{array}{c} \bullet \quad y_{n_t} \\ \vdots \\ \bullet \quad y_{n_2} \\ | \\ \bullet \quad y_{n_1} \end{array} \right).$$

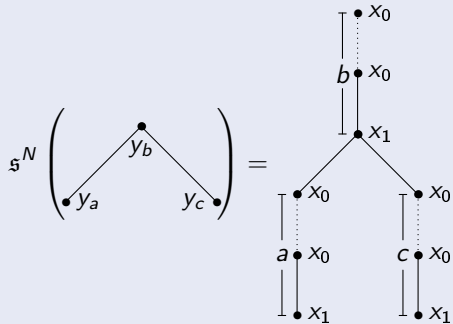
Answer (Manchon, commute, non-nature)

$$\bullet \quad \mathfrak{s}^T := \ell_{\mathcal{X}} \circ \mathfrak{s} \circ \mathfrak{a}_{\mathcal{Y}}$$

$$\begin{array}{ccc} \mathcal{H}_{BCK}^{\mathcal{Y}} & \xrightarrow{\mathfrak{s}^T} & \mathcal{H}_{BCK}^{\mathcal{X}} \\ \mathfrak{a}_{\mathcal{Y}} \downarrow & & \uparrow \ell_{\mathcal{X}} \quad \downarrow \mathfrak{a}_{\mathcal{X}} \\ \mathbb{Q}\langle \mathcal{Y} \rangle & \xrightarrow{\mathfrak{s}} & \mathbb{Q}\langle \mathcal{X} \rangle \end{array}$$

Clavier's attempt

Example (natural map)



$$\ell_X \circ s \circ \alpha_Y(\bullet y_n) = \ell_X \circ s(y_n) = \ell_X(x_1 x_0^{n-1}) = n \bullet x_0 = s^N(\bullet y_n)$$

The diagram shows a forest structure corresponding to the equation. It consists of a single tree with root x_1 and a single child x_0 . Above x_0 is a node x_0 , and above that is a node x_0 . Dotted lines connect x_0 to the top x_0 , and the top x_0 to x_0 . Solid lines connect x_1 to x_0 .

Clavier's attempt

Theorem (Clavier, 2020)

Let F be a forest in $\mathcal{H}_{BCK}^{\mathcal{Y}}$. If $\zeta(F)$ converges, then we have

$$\zeta(\mathfrak{s}^N(F)) \leq \zeta(F).$$

Furthermore, the equality holds if, and only if, F is a ladder forest.

Answer (Clavier, non-commute, nature)

- $\mathfrak{s}^T := \mathfrak{s}^N$

$$\begin{array}{ccc}
 \mathcal{H}_{BCK}^{\mathcal{Y}} & \xrightarrow{\mathfrak{s}^T} & \mathcal{H}_{BCK}^{\mathcal{X}} \\
 \text{a}_{\mathcal{Y}} \downarrow & & \downarrow \text{a}_{\mathcal{X}} \\
 \mathbb{Q}\langle \mathcal{Y} \rangle & \xrightarrow{\mathfrak{s}} & \mathbb{Q}\langle \mathcal{X} \rangle
 \end{array}$$

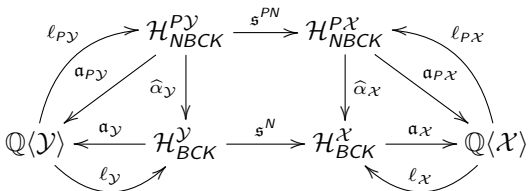
Lift of Manchon's question

Based on Foissy's work, we generalize our setting to the case of planar rooted trees.

Definition

The natural projection from NBCK Hopf algebra $\mathcal{H}_{NBCK}^{PD} = \mathbb{Q}\langle \mathcal{T}^{PD} \rangle$ to BCK Hopf algebra $\mathcal{H}_{BCK}^D = \mathbb{Q}[\mathcal{T}^D]$ by removing the total order relation is denoted by $\hat{\alpha}_D$, which is an algebra morphism.

The lifting map $\mathfrak{a}_{P\mathcal{X}}, \ell_{P\mathcal{X}}, s^{PN}, \mathfrak{a}^{P\mathcal{Y}}, \ell_{P\mathcal{Y}}$ are defined by such that the following diagram commutative.



Main result

Main theorem

Definition

- Y : $\mathcal{Y}$ -decorated planar rooted tree
- δ_Y : decoration map, a map from $V(Y)$ to $\mathcal{Y}$
- Y^e : error term of Y , an element in $\mathcal{H}_{NBCK}^{P\mathcal{Y}}$
- $pr(Y)$: process tree of Y , a $\mathcal{H}_{NBCK}^{P\mathcal{Y}}$ -decorated planar rooted tree

The linear map $\phi : \mathcal{H}_{NBCK}^{P\mathcal{Y}} \rightarrow \mathcal{H}_{NBCK}^{P\mathcal{Y}}$ is defined by

$$\phi(Y) = Y + \sum_{v \in V(pr(Y))} (\delta_{pr(Y)}(v))^e.$$

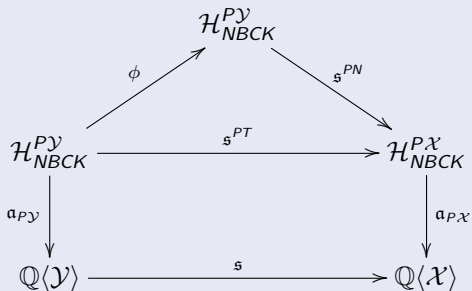
The map $\mathfrak{s}^{PT}$ is defined by

$$\mathfrak{s}^{PT}(Y) = \mathfrak{s}^{PN} \circ \phi(Y).$$

Main theorem

Theorem (F.)

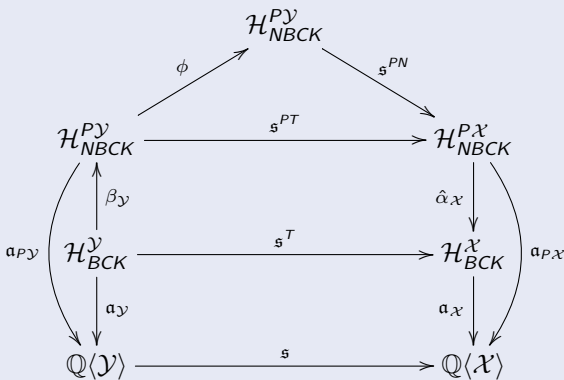
The following diagram is commutative.



Answer to Manchon's question

Corollary (F.)

Let β_Y be the section of $\hat{\alpha}_Y$. The following diagram is commutative.



References

References

- Fan, Ku-Yu, *A map between arborifications of multiple zeta values*, arXiv:2508.20387 [math.NT], 2025, doi:10.48550/arXiv.2508.20387, <https://arxiv.org/abs/2508.20387>

Thank you for your attention!